

CSIT305 Emerging Information Technologies and their Applications

Group Project

Session 3, 2026

DUE DATE	Softcopy to be uploaded to UOW Moodle, 25 th August 2026k 12 by 11:55pm.
WEIGHT OF ASSIGNMENT	30% of total assessment
TYPE OF ASSIGNMENT	Group, Minimum in a group is 4 and maximum is 5.
REPORT LENGTH	4000 – 6000 words
INSTRUCTIONS	The report needs to be neatly typed and should present a professional appearance (use double-spacing and 12 font).
DELIVERABLES	• Project Report in pdf - 25% (submit your completed project report to Project Report Submission folder on Moodle). • Project Presentation in ppt/pptx/pdf - 5% (Presentations would be in the last tutorial.)

Background

In this project, you will work in a team to design a smart city IoT system that will be deployed in the cloud. Your system should integrate IoT devices, networking, multi-modal databases, and cloud-based software to address a real-world urban challenge (e.g., traffic management, waste monitoring, environmental sensing, public safety, smart energy).

You will:

- Identify the key system functions your IoT solution will support.
- Describe the communication protocols, data storage, and processing layers.
- Propose an architecture that incorporates edge computing, cloud services, AI-enabled capabilities, and robust security.
- Present a comprehensive design report covering all required elements in “The Tasks” section.

You are expected to apply your knowledge of multi-modal databases, software engineering, and networking technologies to propose a solution that is technically sound, innovative, and relevant to real-world smart city needs.

The Tasks

System Overview Description (25 marks):

System Goal (10 marks) – Clearly define the overall goal of your IoT-based smart city system, the real-world need it addresses, and the target stakeholders. Conduct brief research into existing systems in the same domain and identify how your proposed system improves upon or differs from them in terms of functionality, usability, performance, or innovation.

Key functions (15 marks) – Identify and describe at least three core functions of your IoT system (e.g., real-time monitoring, automated response, predictive analytics).

For each function, explain:

- Its significance in achieving the system’s goal.
- How it uses IoT sensing, data processing, or cloud-based intelligence to deliver value.

For example: If your system is an autonomous traffic management system, a key function might be “Real-time congestion detection and adaptive signal control” using camera sensors and edge analytics.

Use of Multi-modal Database (20 marks):

Dataset Storage (10 marks) – Describe at least three types of data your system will handle, such as:

- Spatial data (e.g., maps, geolocation)
- Graph data (e.g., sensor connectivity, traffic networks)

- Time-series data (e.g., sensor readings over time)
- Document or log data (e.g., system events, user reports)

Explain how these will be stored in a multi-modal database. Justify your chosen data models (graph, document, key-value, relational) in terms of performance, scalability, and query needs.

Dataset Query Processing (10 marks) – Describe the queries and analytics your system will support, linked to your key functions.

Examples:

- Spatial queries – Identify all sensors in a given disaster zone.
- Graph queries – Detect critical points in a traffic network.
- Time-series queries – Find abnormal temperature spikes over the last 24 hours.
- Text queries – Search for incident reports in logs.

Use of Software (20 marks):

Software Design and Architecture (10 marks) - Present your system's software architecture. Specify the following:

- Architecture style (layered, microservices, event-driven)
- How IoT device firmware, edge analytics, and cloud services interact
- Deployment strategy in the cloud (containerization, serverless functions, load balancing)
- Scalability and fault tolerance mechanisms
- Include a system diagram showing data flow from IoT devices to cloud services and user-facing applications.

Application of AI for Software Engineering (10 marks) - Explain how artificial intelligence is used to support or enhance the software development process or system capabilities. Describe the models or tools you use and explain how they contribute to the system goal. For example, the predictive maintenance for IoT devices and the anomaly detection for sensor data.

Use of Networking (20 marks):

Networking Infrastructure and Integration (10 marks) – Describe how your IoT system's networking layer enables reliable communication and data exchange between devices, edge nodes, and the cloud. Your description should cover the types of IoT devices used (e.g., environmental sensors, cameras, actuators) and the communication protocols selected (such as HTTP/HTTPS), explaining why they are suitable for your application's bandwidth, latency, and reliability requirements. Identify the network technologies employed (e.g., Wi-Fi or 4G/5G) and how they support coverage, energy efficiency, and scalability.

Advanced Technologies and Security (10 marks) – Discuss how advanced networking technologies such as digital twin, blockchain, and smart contracts are applied in your system. For example, explain how you integrate digital twins for simulation and real-time visualization of the physical environment. Additionally, outline how your system protects itself from emerging cyber-attacks.

Professionalism of report (15 marks):

Clarity and Grammar (5 marks) - Spelling, grammar, presentation, layout etc.

Referencing (5 marks) - Intext and reference list. Please use the UOW-Harvard style.

Visual Aids (5 marks) - Use diagrams, charts, tables, and architecture illustrations to support your explanations. All visuals should be labelled, referenced in the text, and easy to interpret.

Note that:

Your report must be your own work. Plagiarism or excessive reliance on AI-generated text will be penalized. If you are using any AI tools in this report, you must note how you use AI tools transparently.

Group Project Presentation Instruction

- Each group (4-5 members) will give a 12-minute presentation on their research report in the last one lab, followed by 3 minutes for questions and feedback. Please manage your time carefully. You will be stopped after 12 minutes of presenting, so plan accordingly. The demonstrator will give you a warning as you approach the time limit.
- Your demonstrator will allocate the date for your presentation. Please prepare the presentation slides and be ready in the date and time allocated to your team.
- The team must be punctual and ready to present at the scheduled time. If your team misses the allocated slot, there may not be another opportunity to present.
- Attendance is required for all students during the scheduled lab session, even if you are not presenting. All team members must participate in and contribute to the presentation. It is up to each group to decide the order, timing, and task allocation.
- Speak clearly and avoid reading directly from a script during your presentation. Otherwise, your team may receive a score deduction.
- Teamwork is very important for this project. Please meet regularly with all your team members (e.g., through Zoom or Webex) and ensure that everyone contributes towards the report and the presentation.
- Please discuss with your lecturer if your team is facing any problems.